

CS 4375 Project Report

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Project Topic:

This project focuses on stock market time series prediction using Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) models. The goal is to use historical stock price data to forecast future values and compare the effectiveness of these two models in capturing temporal patterns over time.

Technique/Algorithm

The main techniques used in this project will be Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks. Both models are designed for sequence data and can use previous time steps to make predictions about future values. A basic RNN can capture short-term temporal patterns, while LSTM is better at learning long-term dependencies and reducing the vanishing gradient problem. In this project, we plan to implement the main logic of these models ourselves instead of relying entirely on built-in high-level libraries, in order to better understand how they work internally. We will train both models on stock market data and compare their prediction accuracy and behavior.

Dataset Details

We will be using a dataset containing stock market data from Yahoo Finance, covering daily prices from April 2018 to March 2023. The dataset contains 7 features and 1258 instances, representing the daily trading records across the years. As a time series dataset, it does not follow a simple independent statistical distribution, since values are temporally dependent and strongly influenced by previous time steps.

Coding Language / Technique

The project will be implemented in Python. We plan to use libraries such as NumPy and Pandas for data preprocessing and handling, and Matplotlib for visualization of results. The main RNN and LSTM model logic will be implemented by our team as much as possible, while helper libraries may be used for loading data, preprocessing, and evaluation. We will also use common evaluation metrics such as Mean Squared Error (MSE) or Root Mean Squared Error (RMSE) to measure prediction performance.

Preliminary Results

At this stage, we have selected the project topic and are researching implementations of a LSTM model for our time series application. We are also currently planning the preprocessing steps needed for sequential input data for our model.